

The logo of K. J. Somaiya College of Engineering is a shield-shaped emblem. It features a central illustration of a building with a dome, flanked by two smaller structures. Below the building, there is a banner with the text 'K. J. SOMAIYA COLLEGE OF ENGINEERING'. The shield is divided into four quadrants, each containing a different symbol: a gear, a wrench, a screwdriver, and a pair of pliers. The entire logo is set against a light blue background.

Experiment No.: 04

Title: Programing with Jenkins

Batch: B-1

Roll No.: 16010422234

Experiment No.: 04

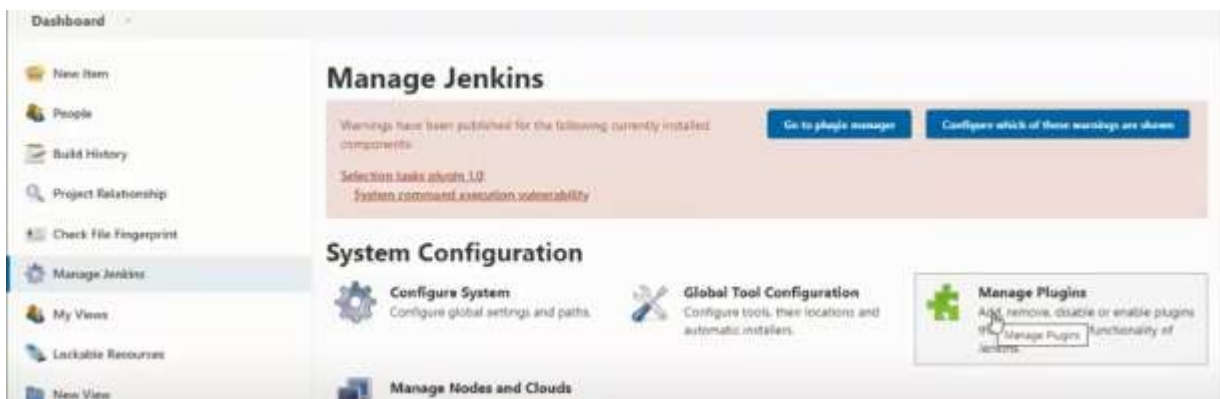
Aim: Executing Java program with Jenkins

Resources needed: Jenkins, Java

Theory:

Manage plugins: To install plugins for integration with Jenkins.

Global Tool Configuration: To Configure plugins for integration with Jenkins



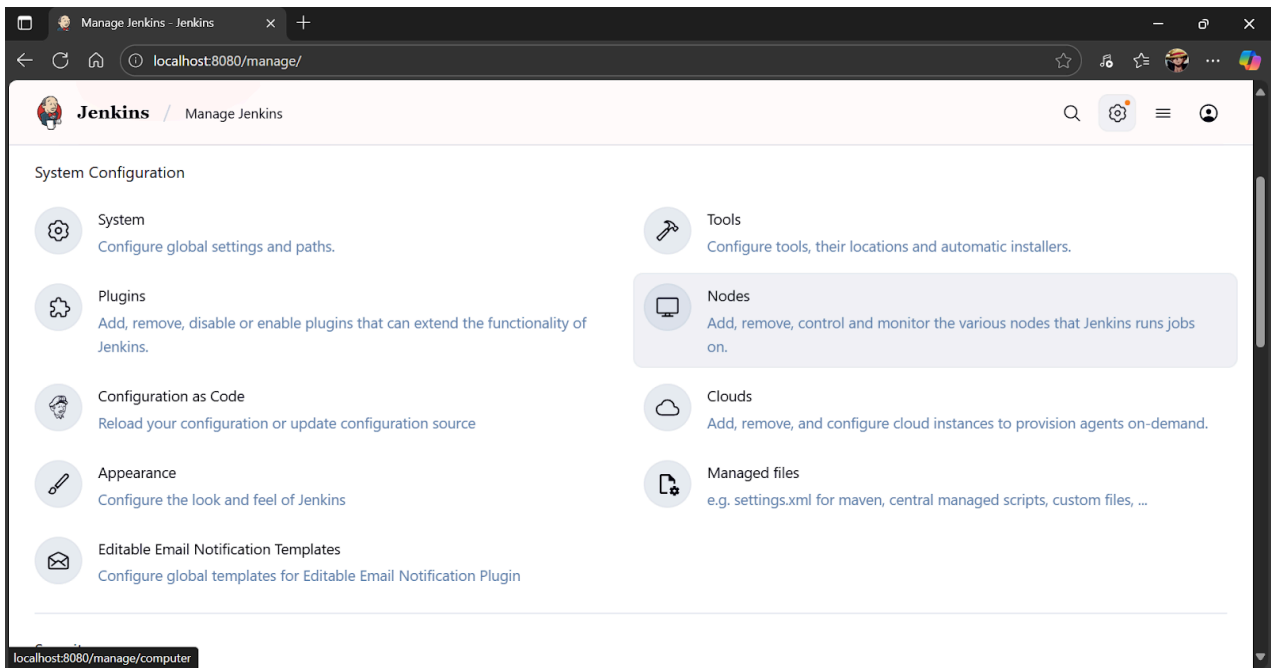
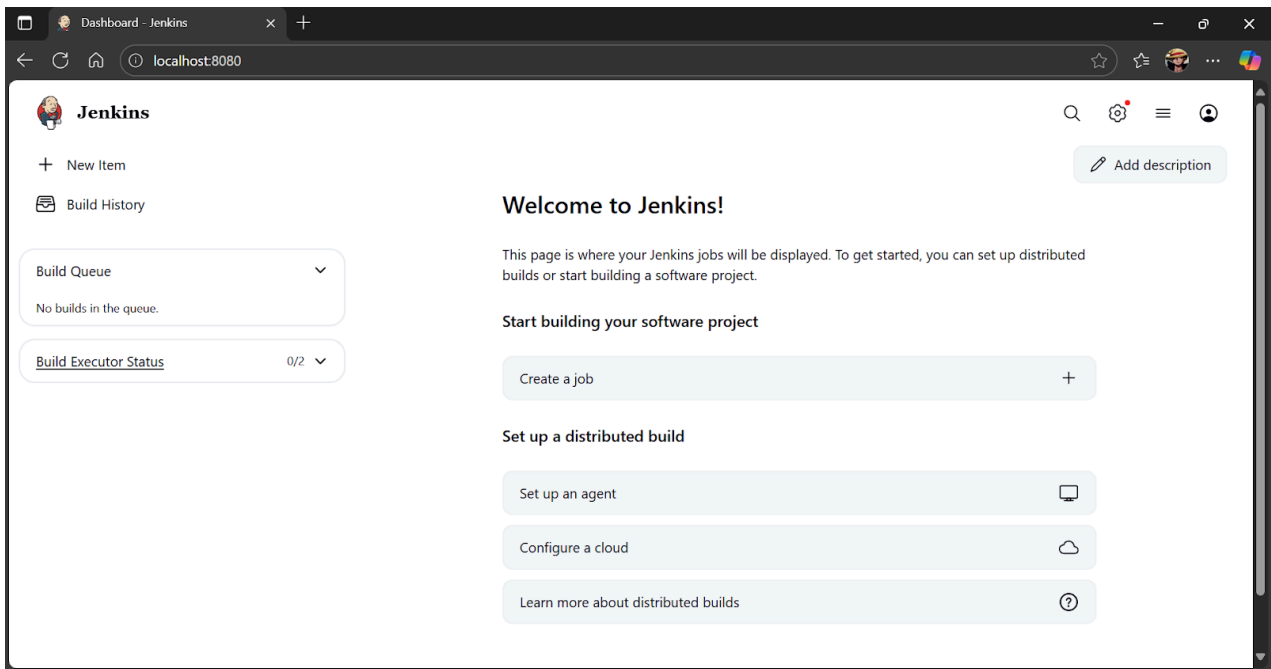
Procedure:

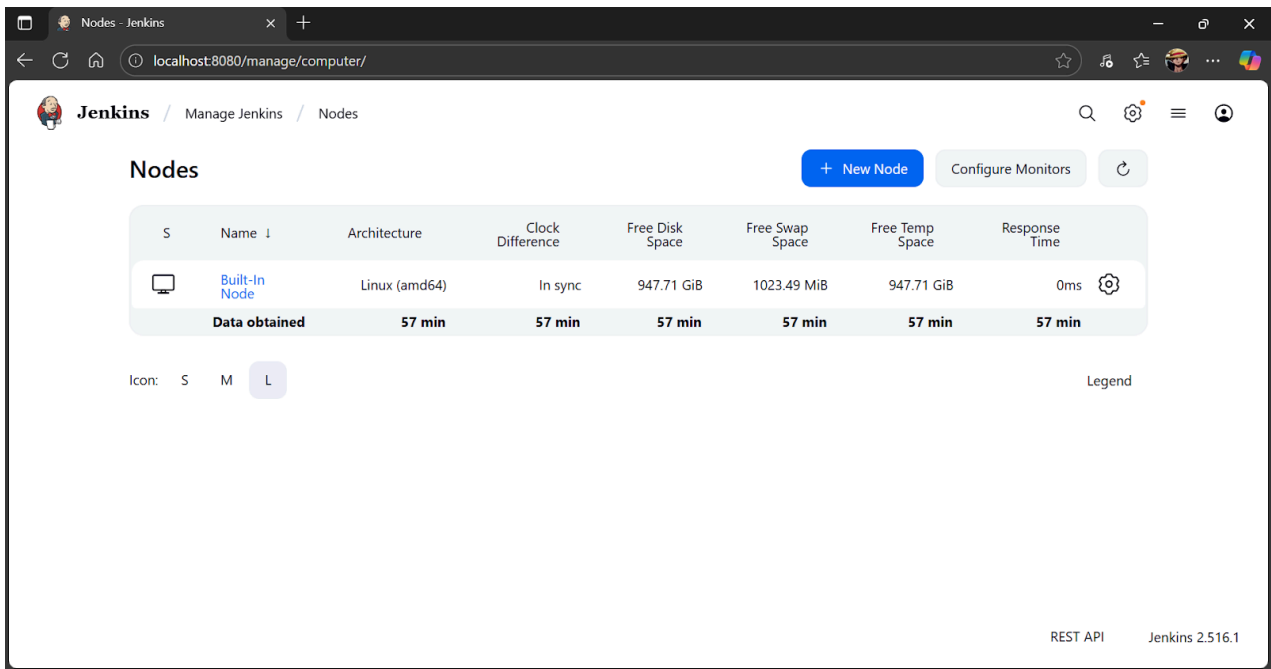
Java parametrized Programming [Explicit]

1. Create New item-> freestyle project this project is parametrized
 2. Set two string parameter num1 and num2
 3. Select build tab->window batch command, write command and build
-

Results: (Document with screenshots)

1. All steps to execute Explicit java parametrized program.



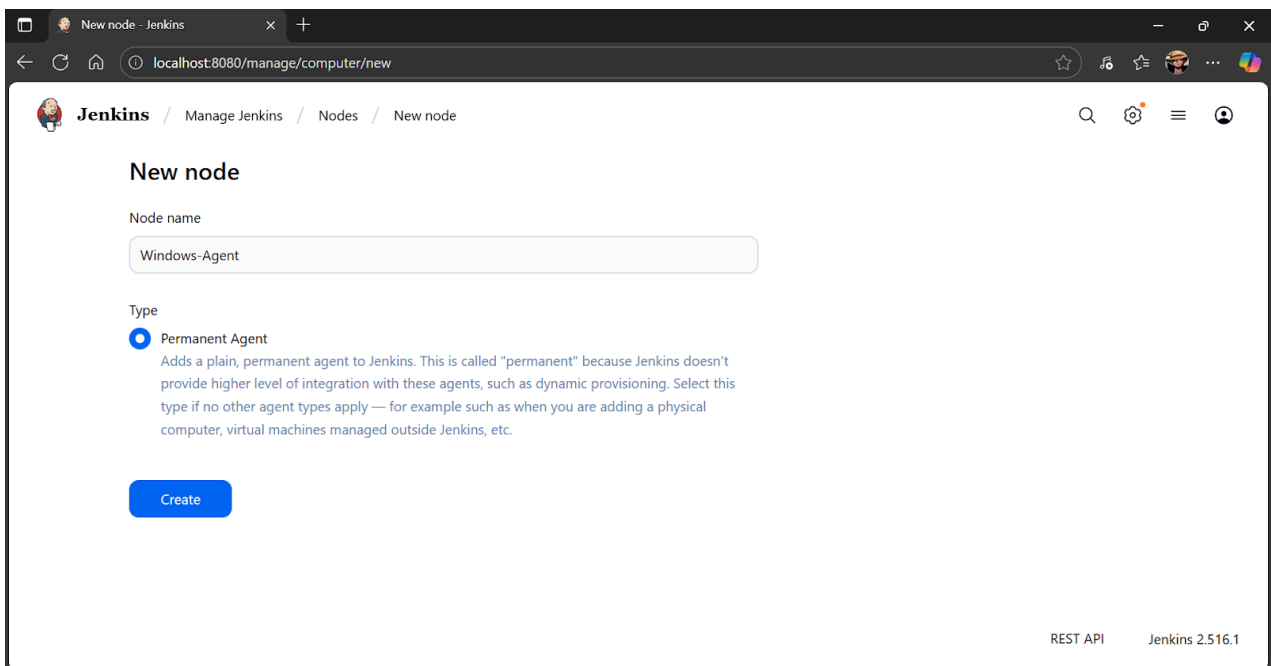


The screenshot shows the Jenkins 'Nodes' management page. The breadcrumb trail is 'Jenkins / Manage Jenkins / Nodes'. The page title is 'Nodes'. There are two buttons: '+ New Node' and 'Configure Monitors'. A table lists the nodes with columns: S, Name, Architecture, Clock Difference, Free Disk Space, Free Swap Space, Free Temp Space, and Response Time. One node is listed: 'Built-In Node' with architecture 'Linux (amd64)', clock difference 'In sync', and various resource metrics. Below the table, there is a 'Data obtained' row showing '57 min' for several metrics. At the bottom, there is a 'Legend' section with icons for S, M, and L. The footer shows 'REST API' and 'Jenkins 2.516.1'.

S	Name	Architecture	Clock Difference	Free Disk Space	Free Swap Space	Free Temp Space	Response Time
	Built-In Node	Linux (amd64)	In sync	947.71 GiB	1023.49 MiB	947.71 GiB	0ms
Data obtained		57 min	57 min	57 min	57 min	57 min	57 min

Icon: S M L Legend

REST API Jenkins 2.516.1



The screenshot shows the Jenkins 'New node' page. The breadcrumb trail is 'Jenkins / Manage Jenkins / Nodes / New node'. The page title is 'New node'. There is a 'Node name' field with the value 'Windows-Agent'. Below it, there is a 'Type' section with a radio button selected for 'Permanent Agent'. A description for 'Permanent Agent' is provided: 'Adds a plain, permanent agent to Jenkins. This is called "permanent" because Jenkins doesn't provide higher level of integration with these agents, such as dynamic provisioning. Select this type if no other agent types apply — for example such as when you are adding a physical computer, virtual machines managed outside Jenkins, etc.' At the bottom, there is a 'Create' button. The footer shows 'REST API' and 'Jenkins 2.516.1'.

Node name

Windows-Agent

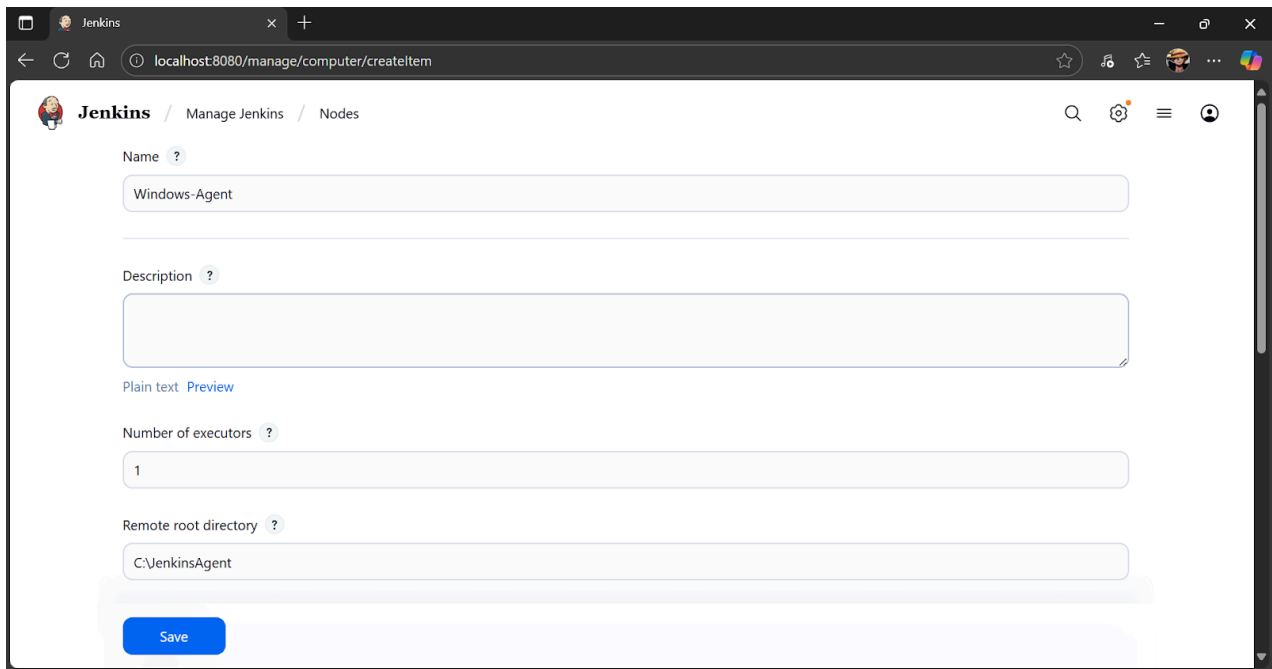
Type

☒ Permanent Agent

Adds a plain, permanent agent to Jenkins. This is called "permanent" because Jenkins doesn't provide higher level of integration with these agents, such as dynamic provisioning. Select this type if no other agent types apply — for example such as when you are adding a physical computer, virtual machines managed outside Jenkins, etc.

Create

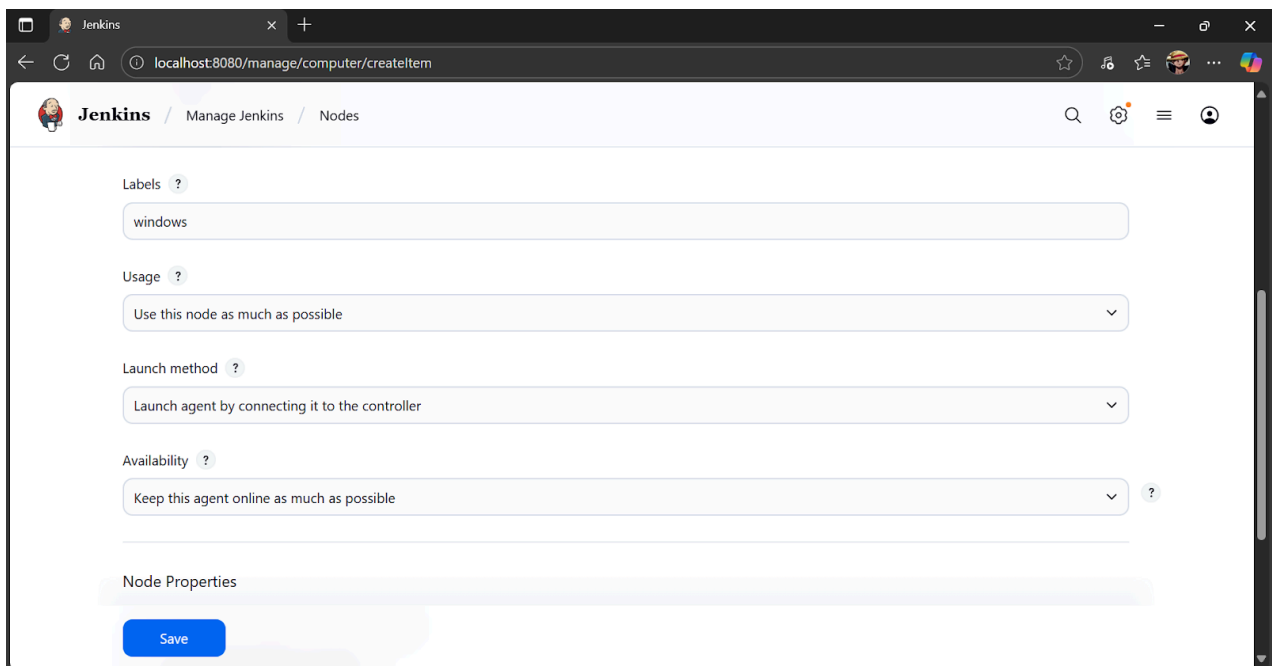
REST API Jenkins 2.516.1



The screenshot shows the Jenkins 'createItem' page for a new node. The browser address bar shows 'localhost:8080/manage/computer/createItem'. The page title is 'Jenkins / Manage Jenkins / Nodes'. The form contains the following fields:

- Name**: 'Windows-Agent'
- Description**: (empty text area)
- Plain text**: [Preview](#)
- Number of executors**: '1'
- Remote root directory**: 'C:\JenkinsAgent'

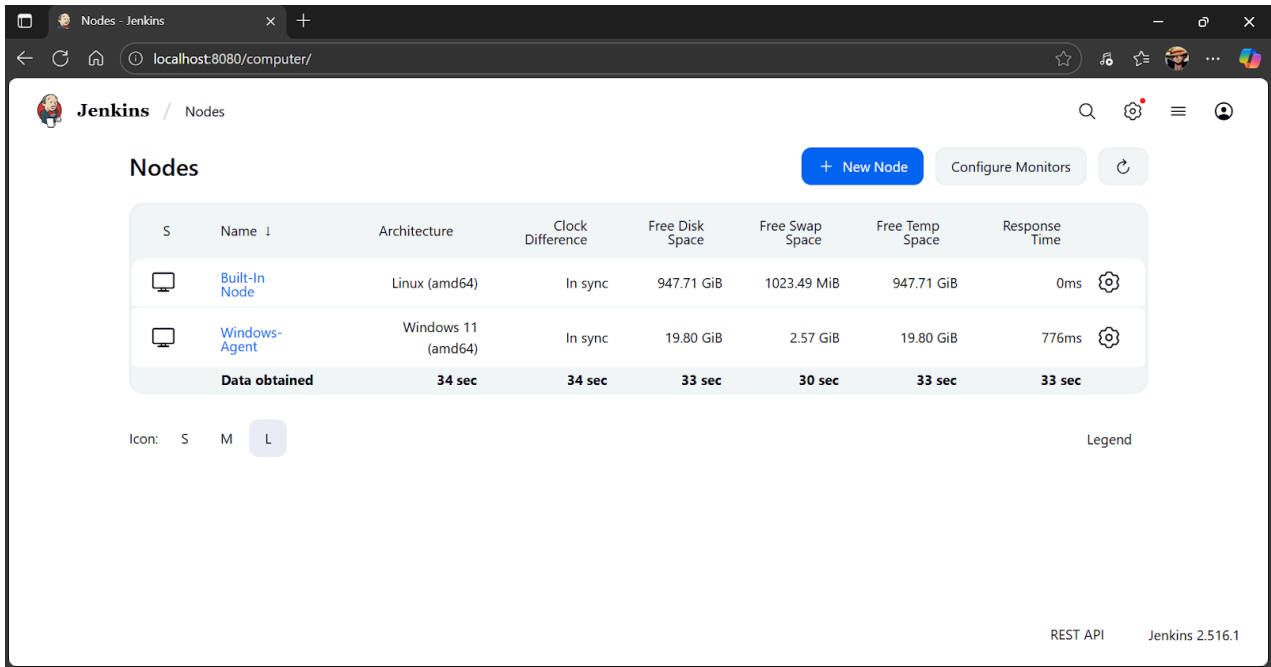
A blue 'Save' button is located at the bottom of the form.



The screenshot shows the Jenkins 'createItem' page for a new node, displaying configuration options. The browser address bar shows 'localhost:8080/manage/computer/createItem'. The page title is 'Jenkins / Manage Jenkins / Nodes'. The form contains the following fields:

- Labels**: 'windows'
- Usage**: 'Use this node as much as possible' (dropdown menu)
- Launch method**: 'Launch agent by connecting it to the controller' (dropdown menu)
- Availability**: 'Keep this agent online as much as possible' (dropdown menu)

A blue 'Save' button is located at the bottom of the form.

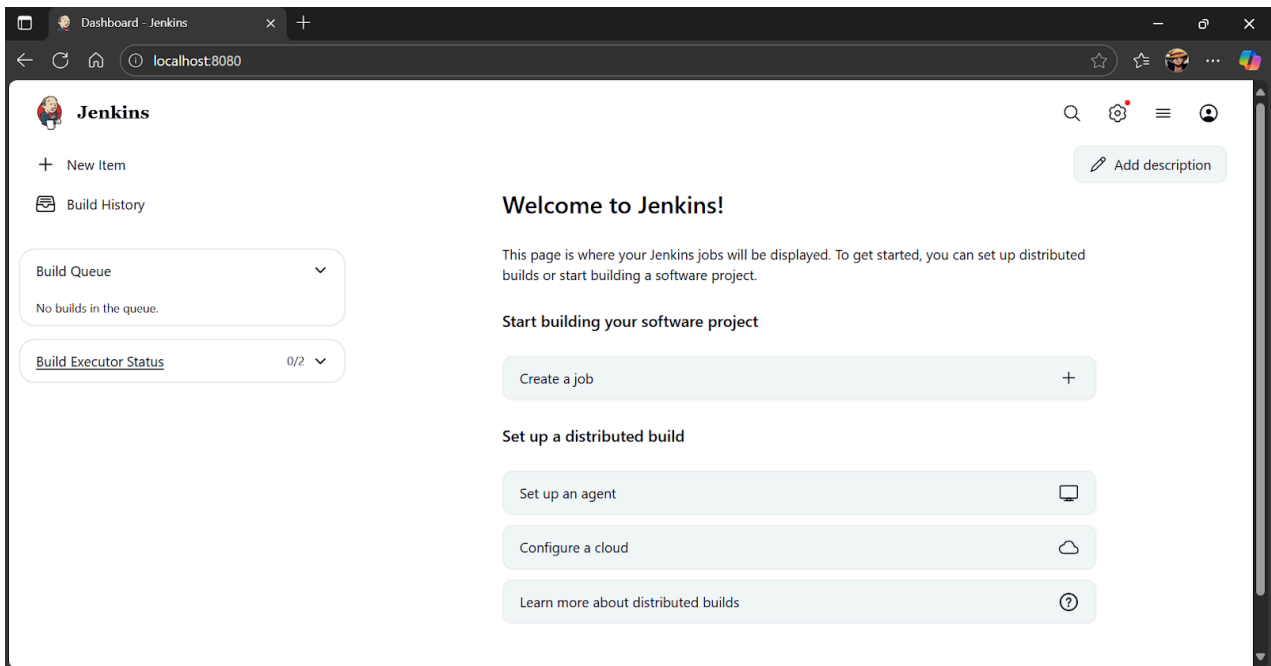


The screenshot shows the Jenkins 'Nodes' page in a web browser. The page title is 'Nodes' and the URL is 'localhost:8080/computer/'. The page features a table of nodes and a 'New Node' button.

S	Name ↓	Architecture	Clock Difference	Free Disk Space	Free Swap Space	Free Temp Space	Response Time
	Built-In Node	Linux (amd64)	In sync	947.71 GiB	1023.49 MiB	947.71 GiB	0ms
	Windows-Agent	Windows 11 (amd64)	In sync	19.80 GiB	2.57 GiB	19.80 GiB	776ms
Data obtained		34 sec	34 sec	33 sec	30 sec	33 sec	33 sec

Icon: S M L Legend

REST API Jenkins 2.516.1



The screenshot shows the Jenkins 'Dashboard' page in a web browser. The page title is 'Dashboard - Jenkins' and the URL is 'localhost:8080'. The page features a 'Welcome to Jenkins!' message and several buttons for creating jobs and setting up distributed builds.

Welcome to Jenkins!

This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project.

Start building your software project

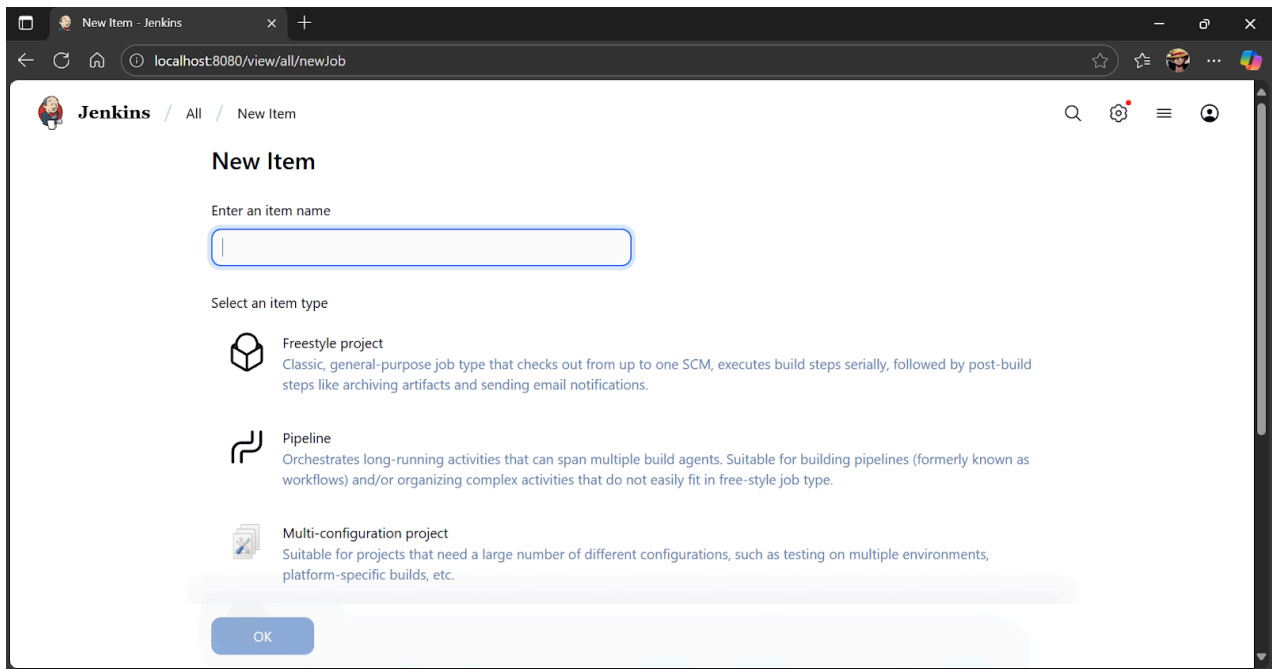
Create a job +

Set up a distributed build

Set up an agent 

Configure a cloud 

Learn more about distributed builds 



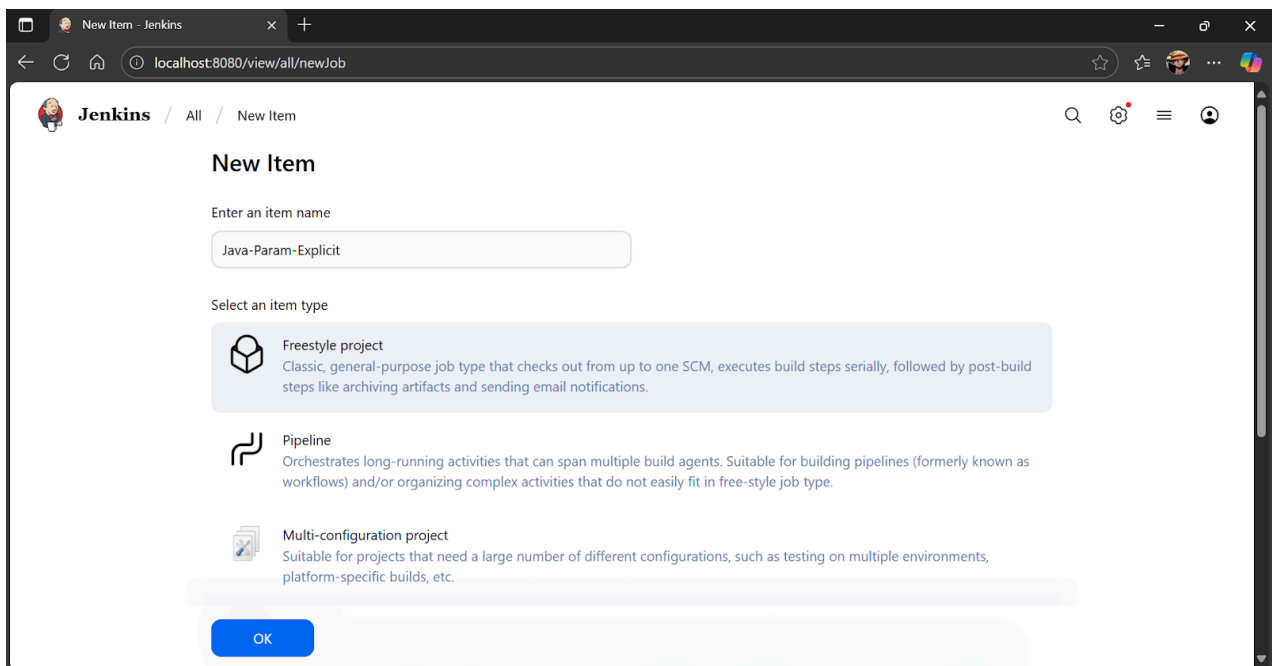
The screenshot shows the Jenkins 'New Item' page in a web browser. The browser's address bar displays 'localhost:8080/view/all/newJob'. The Jenkins header includes the logo, 'Jenkins', and navigation links 'All' and 'New Item'. The main heading is 'New Item'. Below it, the label 'Enter an item name' is followed by an empty text input field. Under the heading 'Select an item type', three options are listed: 'Freestyle project' (with a cube icon), 'Pipeline' (with a flow icon), and 'Multi-configuration project' (with a document icon). Each option includes a brief description. At the bottom left, there is a blue 'OK' button.

Enter an item name

Select an item type

- Freestyle project**
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
- Pipeline**
Orchestrates long-running activities that can span multiple build agents. Suitable for building pipelines (formerly known as workflows) and/or organizing complex activities that do not easily fit in free-style job type.
- Multi-configuration project**
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

OK



This screenshot is identical to the one above, but the text input field for the item name now contains the text 'Java-Param-Explicit'.

Enter an item name

Java-Param-Explicit

Select an item type

- Freestyle project**
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
- Pipeline**
Orchestrates long-running activities that can span multiple build agents. Suitable for building pipelines (formerly known as workflows) and/or organizing complex activities that do not easily fit in free-style job type.
- Multi-configuration project**
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

OK

Java-Param-Explicit Config - Jenkins

localhost:8080/job/Java-Param-Explicit/configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

General

Enabled ☒

Description

Plain text [Preview](#)

☐ Discard old builds ?

☐ GitHub project

GitLab Connection

☐ Use alternative credential

[Save](#) [Apply](#)

Java-Param-Explicit Config - Jenkins

localhost:8080/job/Java-Param-Explicit/configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

General

Enabled ☒

Description

Plain text [Preview](#)

☐ Discard old builds ?

☐ GitHub project

GitLab Connection

☐ Use alternative credential

☐ Permission to Copy Artifact

☒ This project is parameterized ?

[Save](#) [Apply](#)

The screenshot shows the Jenkins configuration page for a job named 'Java-Param-Explicit'. The left sidebar contains a 'Configure' section with a list of categories: General, Source Code Management, Triggers, Environment, Build Steps, and Post-build Actions. The 'General' category is selected. The main content area displays the 'String Parameter' configuration. The 'Name' field is 'num1', the 'Default Value' is '5', and the 'Description' field is empty. Below the description field, there is a 'Trim the string' checkbox which is unchecked. At the bottom, there are 'Save' and 'Apply' buttons.

Configure

General

Source Code Management

Triggers

Environment

Build Steps

Post-build Actions

String Parameter

Name ?

num1

Default Value ?

5

Description ?

Plain text Preview

☐ Trim the string ?

Save Apply

The screenshot shows the Jenkins configuration page for a job named 'Java-Param-Explicit'. The left sidebar contains a 'Configure' section with a list of categories: General, Source Code Management, Triggers, Environment, Build Steps, and Post-build Actions. The 'General' category is selected. The main content area displays the 'String Parameter' configuration. The 'Name' field is 'num2', the 'Default Value' is '7', and the 'Description' field is empty. Below the description field, there is a 'Trim the string' checkbox which is unchecked. At the bottom, there are 'Save' and 'Apply' buttons.

Configure

General

Source Code Management

Triggers

Environment

Build Steps

Post-build Actions

String Parameter

Name ?

num2

Default Value ?

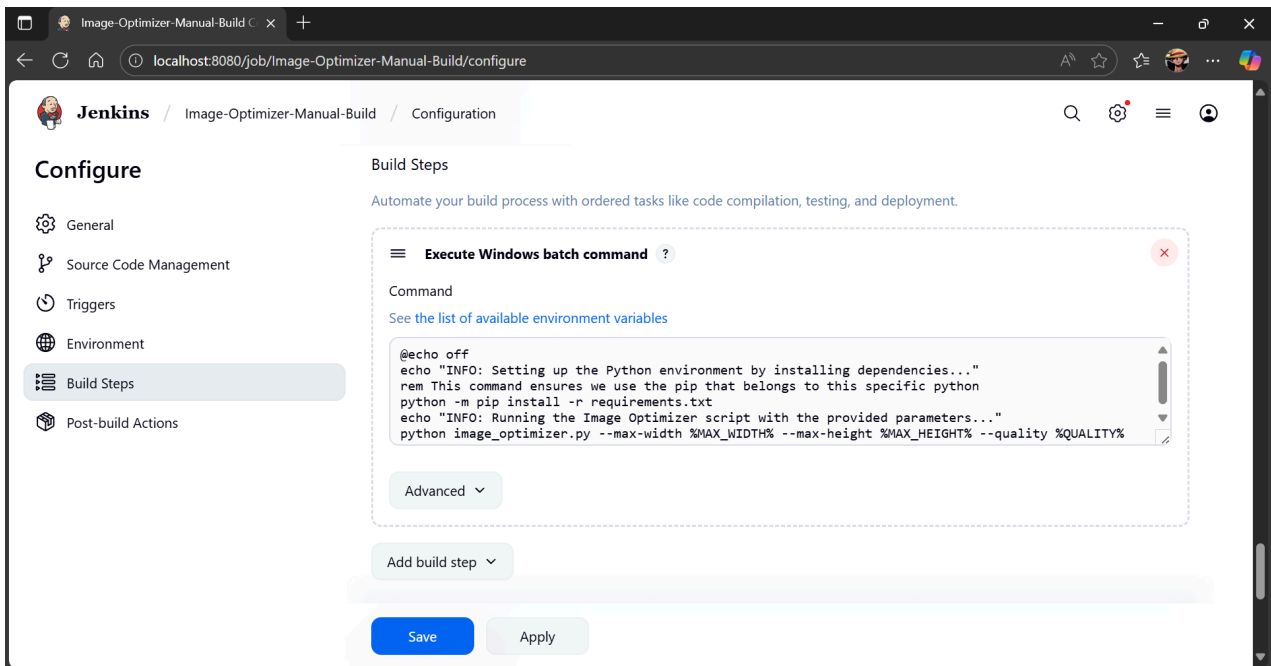
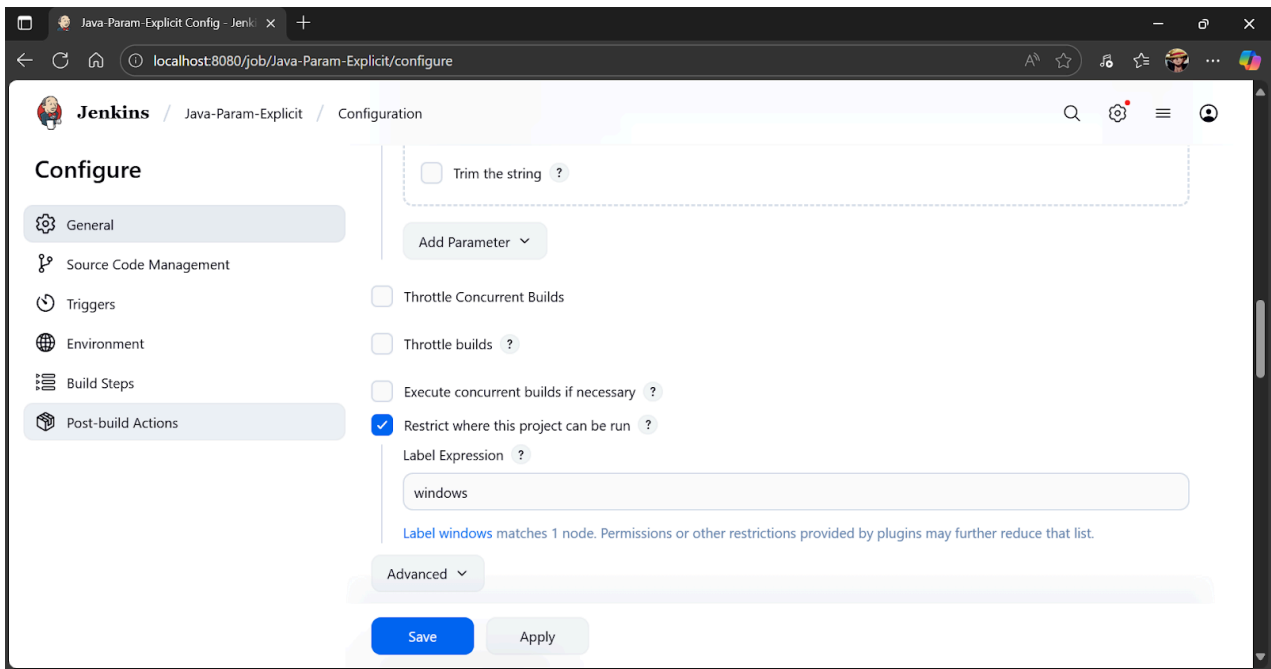
7

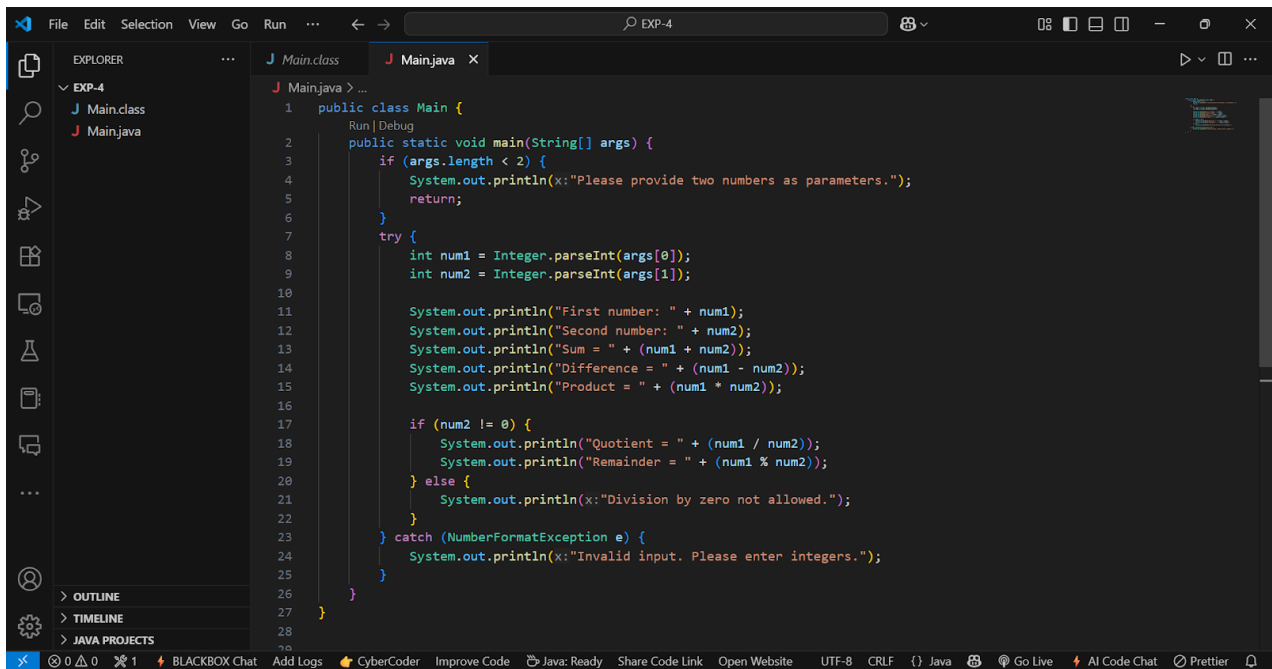
Description ?

Plain text Preview

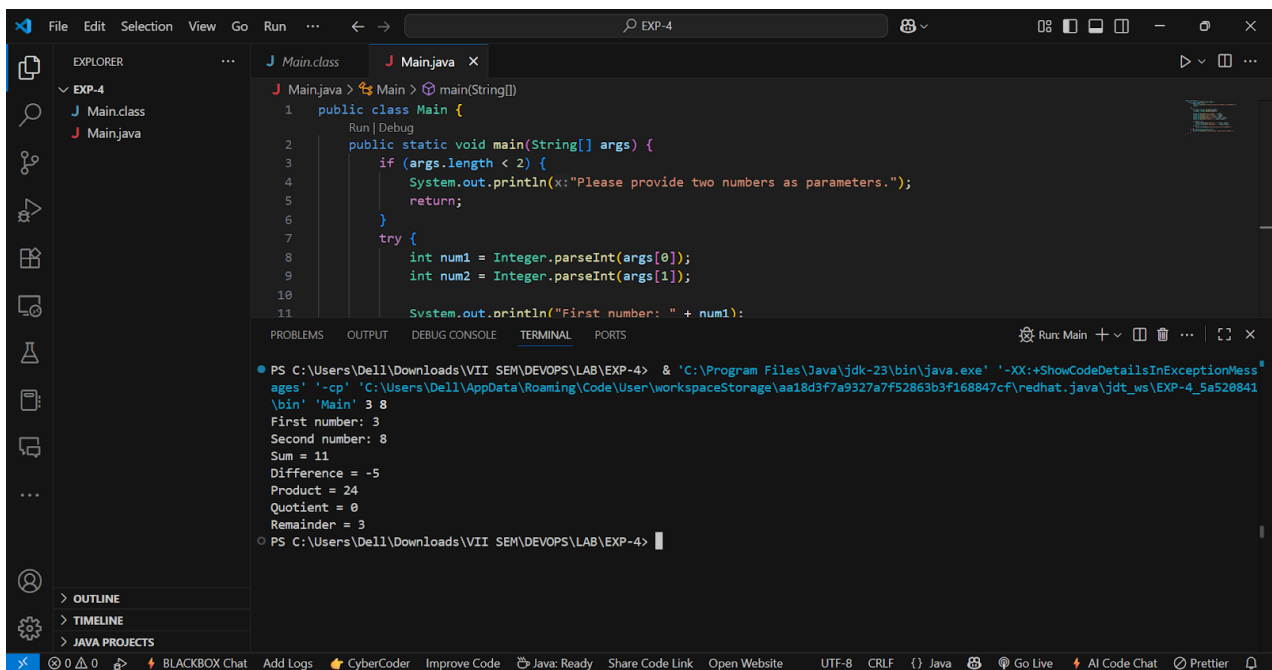
☐ Trim the string ?

Save Apply





```
1 public class Main {
2     public static void main(String[] args) {
3         if (args.length < 2) {
4             System.out.println(x:"Please provide two numbers as parameters.");
5             return;
6         }
7         try {
8             int num1 = Integer.parseInt(args[0]);
9             int num2 = Integer.parseInt(args[1]);
10
11             System.out.println("First number: " + num1);
12             System.out.println("Second number: " + num2);
13             System.out.println("Sum = " + (num1 + num2));
14             System.out.println("Difference = " + (num1 - num2));
15             System.out.println("Product = " + (num1 * num2));
16
17             if (num2 != 0) {
18                 System.out.println("Quotient = " + (num1 / num2));
19                 System.out.println("Remainder = " + (num1 % num2));
20             } else {
21                 System.out.println(x:"Division by zero not allowed.");
22             }
23         } catch (NumberFormatException e) {
24             System.out.println(x:"Invalid input. Please enter integers.");
25         }
26     }
27 }
```



```
PS C:\Users\De11\Downloads\VII SEM\DEVOPS\LAB\EXP-4> & 'C:\Program Files\Java\jdk-23\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\De11\AppData\Roaming\Code\User\workspaceStorage\aa18d3f7a9327a7f52863b3f168847cf\redhat.java\jdt_ws\EXP-4_5a528841\bin\' 'Main' 3 8
First number: 3
Second number: 8
Sum = 11
Difference = -5
Product = 24
Quotient = 0
Remainder = 3
PS C:\Users\De11\Downloads\VII SEM\DEVOPS\LAB\EXP-4>
```

```
Command Prompt
Microsoft Windows [Version 10.0.22621.4317]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Dell>copy "C:\Users\Dell\Downloads\VII SEM\DEVOPS\LAB\EX \Main.java" "C:\JenkinsAgent\workspace\Java-Param-Explicit\Main.j
ava"
1 file(s) copied.

C:\Users\Dell>dir "C:\JenkinsAgent\workspace\Java-Param-Explicit"
Volume in drive C has no label.
Volume Serial Number is E4F5-FAA1

Directory of C:\JenkinsAgent\workspace\Java-Param-Explicit
16-08-2025 16:15 <DIR> .
16-08-2025 16:05 <DIR> ..
16-08-2025 15:36      1,055 Main.java
                1 File(s)      1,055 bytes
                2 Dir(s)  20,243,566,592 bytes free

C:\Users\Dell>
```

Jenkins / Java-Param-Explicit

Project Java-Param-Explicit

This build requires parameters:

num1
3

num2
8

Build Cancel

Builds
No builds

The screenshot shows the Jenkins web interface for a job named 'Java-Param-Explicit'. The browser address bar indicates the URL is `localhost:8080/job/Java-Param-Explicit/`. The Jenkins logo and job name are at the top. On the left, a sidebar contains navigation links: Status (selected), Changes, Workspace, Build with Parameters, Configure, Delete Project, Rename, and Embeddable Build Status. The main content area features a green checkmark icon and the job name 'Java-Param-Explicit', followed by a green checkmark icon and the word 'Permalinks'. Below this, a list of links for the last build (#1) is shown, all indicating a duration of 28 seconds ago. A 'Builds' section on the left includes a search filter and a list of builds, with the first build (#1) at 10:47 AM being visible.

Jenkins / Java-Param-Explicit

Status

Java-Param-Explicit

Permalinks

- Last build (#1), 28 sec ago
- Last stable build (#1), 28 sec ago
- Last successful build (#1), 28 sec ago
- Last completed build (#1), 28 sec ago

Builds

Filter

Today

#1 10:47 AM

The screenshot shows the Jenkins console output for build #1 of the 'Java-Param-Explicit' job. The browser address bar indicates the URL is `localhost:8080/job/Java-Param-Explicit/5/console`. The Jenkins logo and job name are at the top. On the left, a sidebar contains navigation links: Status, Changes, Console Output (selected), Edit Build Information, Delete build '#1', Parameters, Timings, and Embeddable Build Status. The main content area features a green checkmark icon and the words 'Console Output', followed by buttons for Download, Copy, and View as plain text. Below this, the console output text is displayed, showing the build process from user login to successful completion. The output includes the command used to run the Java program and the results of the calculations.

Jenkins / Java-Param-Explicit / #1 / Console Output

Status

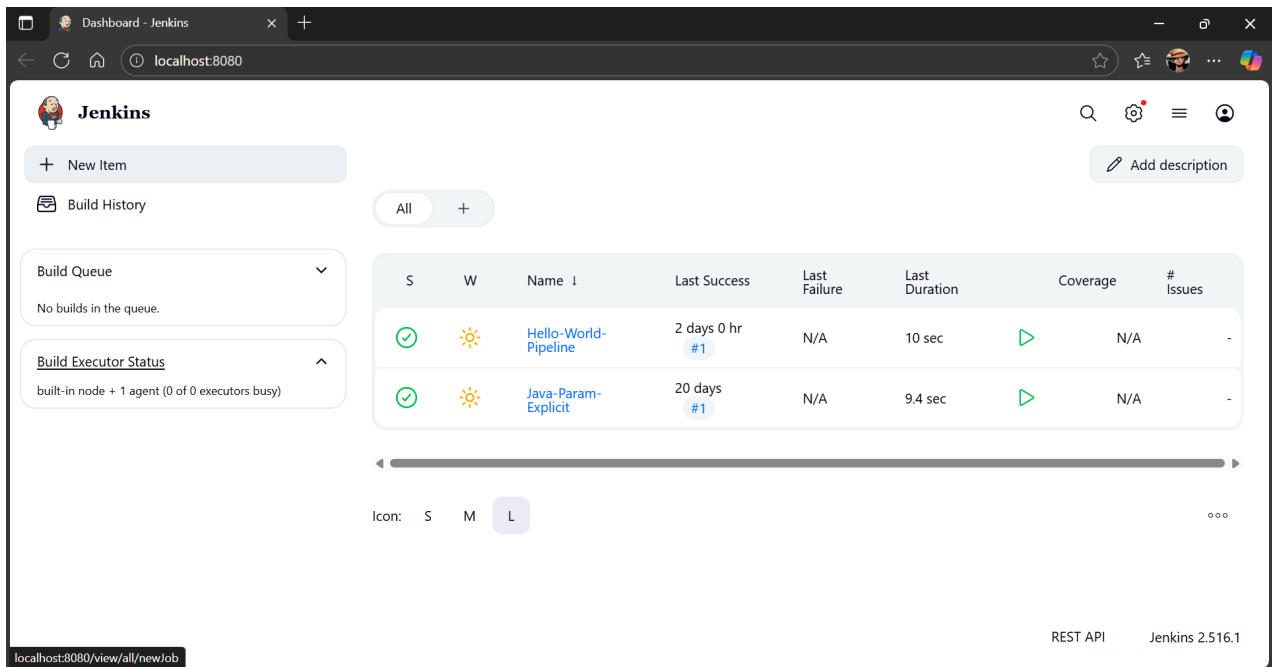
Console Output

Download Copy View as plain text

```
Started by user Chandana Ramesh Galgali
Running as SYSTEM
Building remotely on Windows-Agent (windows) in workspace C:\JenkinsAgent\workspace\Java-Param-Explicit
[Java-Param-Explicit] $ cmd /c call C:\Users\Dell\AppData\Local\Temp\jenkins5034365469701350740.bat
Compiling Java program...
Running Java program with parameters: 3 and 8
First number: 3
Second number: 8
Sum = 11
Difference = -5
Product = 24
Quotient = 0
Remainder = 3
Finished: SUCCESS
```

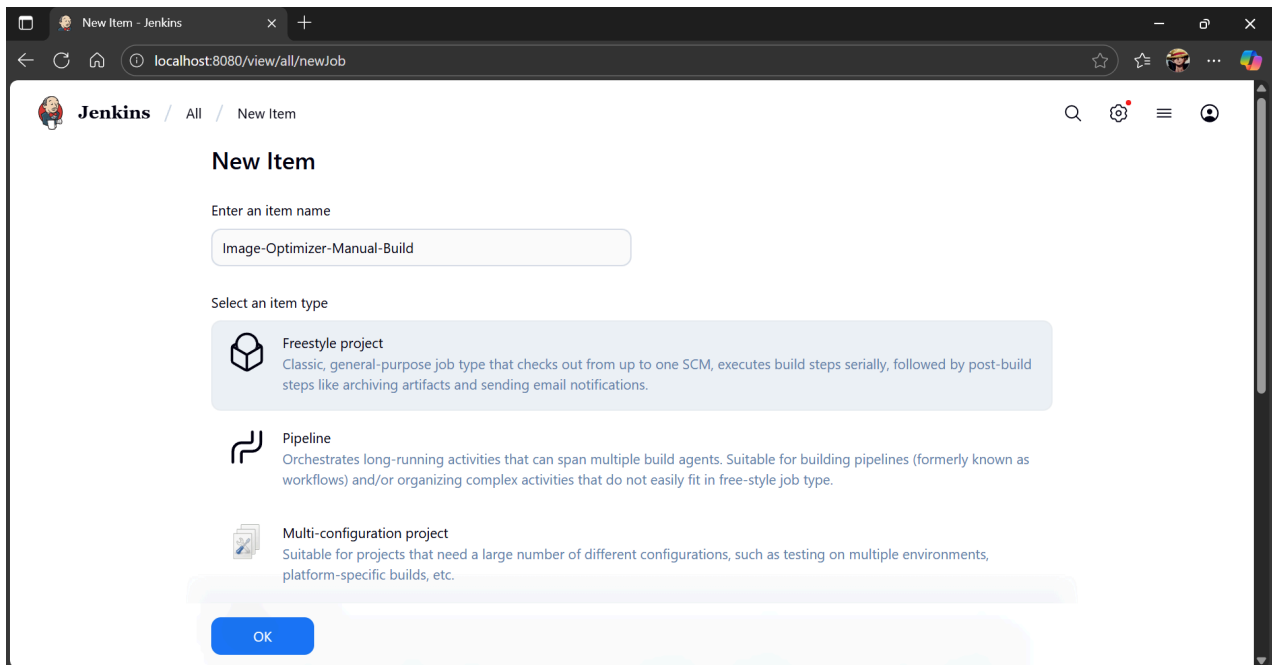
REST API Jenkins 2.516.1

Image-Optimizer-Manual-Build



The screenshot shows the Jenkins Dashboard at localhost:8080. The interface includes a sidebar with 'New Item' and 'Build History' links. The main area displays a table of build history for the 'All' filter. The table has columns for status (S), warnings (W), name, last success, last failure, last duration, coverage, and number of issues. Two builds are listed: 'Hello-World-Pipeline' and 'Java-Param-Explicit', both successful. Below the table is a filter bar with 'Icon' and 'S M L' options. The bottom status bar shows 'localhost:8080/view/all/newJob', 'REST API', and 'Jenkins 2.516.1'.

S	W	Name	Last Success	Last Failure	Last Duration	Coverage	# Issues
✓	☀	Hello-World-Pipeline	2 days 0 hr #1	N/A	10 sec	▶	N/A
✓	☀	Java-Param-Explicit	20 days #1	N/A	9.4 sec	▶	N/A



The screenshot shows the 'New Item' form in Jenkins. The 'Enter an item name' field contains 'Image-Optimizer-Manual-Build'. Under 'Select an item type', three options are available: 'Freestyle project' (Classic, general-purpose job type), 'Pipeline' (Orchestrates long-running activities), and 'Multi-configuration project' (Suitable for projects that need a large number of different configurations). An 'OK' button is at the bottom.

Enter an item name

Image-Optimizer-Manual-Build

Select an item type

- Freestyle project**
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
- Pipeline**
Orchestrates long-running activities that can span multiple build agents. Suitable for building pipelines (formerly known as workflows) and/or organizing complex activities that do not easily fit in free-style job type.
- Multi-configuration project**
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

OK

The screenshot shows the Jenkins Configuration page for a job named 'Image-Optimizer-Manual-Build'. The left sidebar contains a 'Configure' section with a list of tabs: General, Source Code Management, Triggers, Environment, Build Steps, and Post-build Actions. The 'General' tab is selected. At the top, there is a checkbox labeled 'This project is parameterized' which is checked. Below this, a 'String Parameter' configuration block is visible. It has a 'Name' field with the value 'MAX_WIDTH', a 'Default Value' field with the value '1280', and a 'Description' field with the text 'The maximum width for the output images in pixels.' At the bottom of the configuration block, there are links for 'Plain text' and 'Preview'. Below the configuration block, there are 'Save' and 'Apply' buttons.

Jenkins / Image-Optimizer-Manual-Build / Configuration

Configure

☒ This project is parameterized ?

String Parameter ?

Name ?
MAX_WIDTH

Default Value ?
1280

Description ?
The maximum width for the output images in pixels.

Plain text [Preview](#)

[Save](#) [Apply](#)

The screenshot shows the Jenkins Configuration page for a job named 'Image-Optimizer-Manual-Build'. The left sidebar contains a 'Configure' section with a list of tabs: General, Source Code Management, Triggers, Environment, Build Steps, and Post-build Actions. The 'General' tab is selected. At the top, there is a checkbox labeled 'This project is parameterized' which is checked. Below this, a 'String Parameter' configuration block is visible. It has a 'Name' field with the value 'MAX_HEIGHT', a 'Default Value' field with the value '720', and a 'Description' field with the text 'The maximum height for the output images in pixels.' At the bottom of the configuration block, there are links for 'Plain text' and 'Preview'. Below the configuration block, there is a checkbox labeled 'Trim the string' which is unchecked. At the bottom of the page, there are 'Save' and 'Apply' buttons.

Jenkins / Image-Optimizer-Manual-Build / Configuration

Configure

☒ This project is parameterized ?

String Parameter ?

Name ?
MAX_HEIGHT

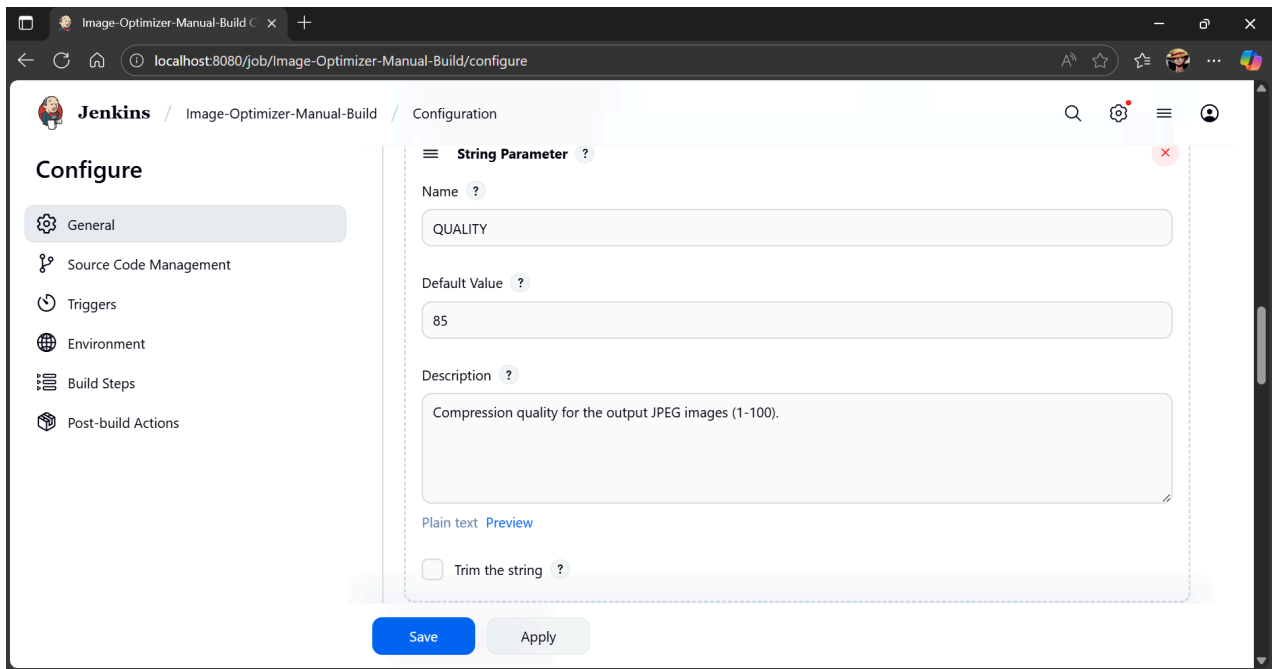
Default Value ?
720

Description ?
The maximum height for the output images in pixels.

Plain text [Preview](#)

☐ Trim the string ?

[Save](#) [Apply](#)



The screenshot shows the Jenkins configuration page for a job named "Image-Optimizer-Manual-Build". The "Configuration" tab is active, and the "String Parameter" section is expanded. The parameter is named "QUALITY" with a default value of "85". The description is "Compression quality for the output JPEG images (1-100)". The "Trim the string" checkbox is unchecked. The "Save" button is highlighted in blue.

Jenkins / Image-Optimizer-Manual-Build / Configuration

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

String Parameter ?

Name ?
QUALITY

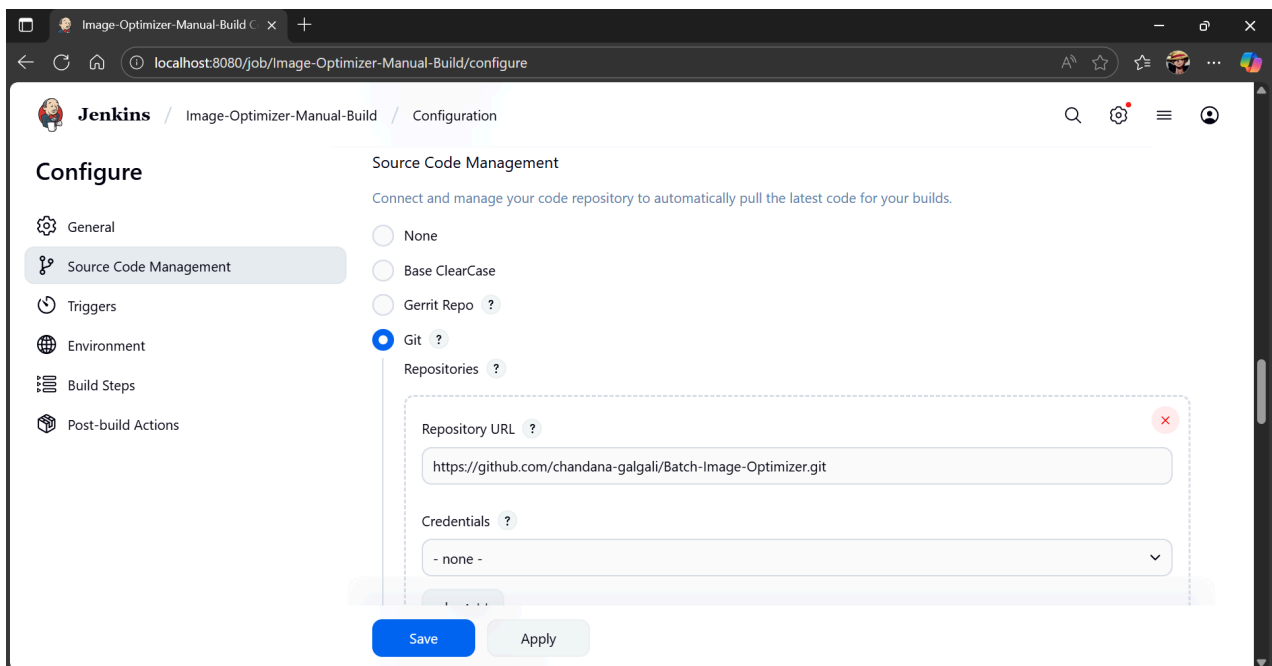
Default Value ?
85

Description ?
Compression quality for the output JPEG images (1-100).

Plain text [Preview](#)

☐ Trim the string ?

[Save](#) [Apply](#)



The screenshot shows the Jenkins configuration page for the same job, but the "Source Code Management" tab is active. The "Git" option is selected under "Connect and manage your code repository". The "Repository URL" is set to "https://github.com/chandana-galgali/Batch-Image-Optimizer.git" and the "Credentials" dropdown is set to "- none -". The "Save" button is highlighted in blue.

Jenkins / Image-Optimizer-Manual-Build / Configuration

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps
- Post-build Actions

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

☐ None

☐ Base ClearCase

☐ Gerrit Repo ?

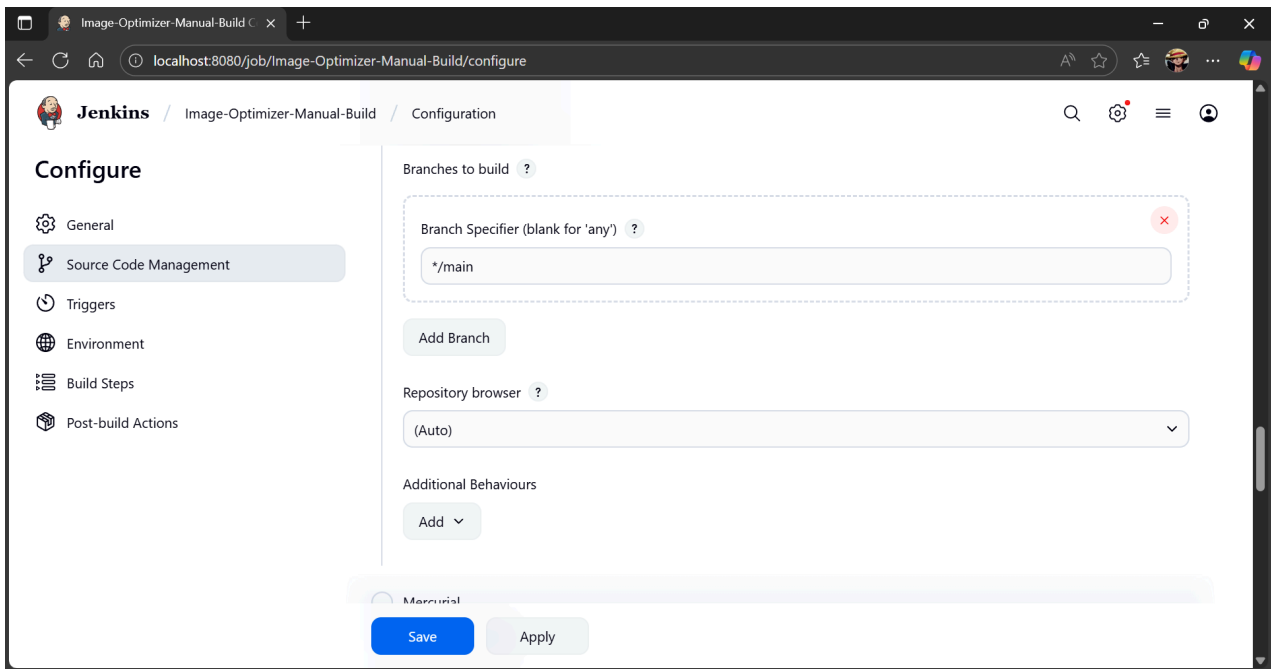
☒ Git ?

Repositories ?

Repository URL ?
https://github.com/chandana-galgali/Batch-Image-Optimizer.git

Credentials ?
- none -

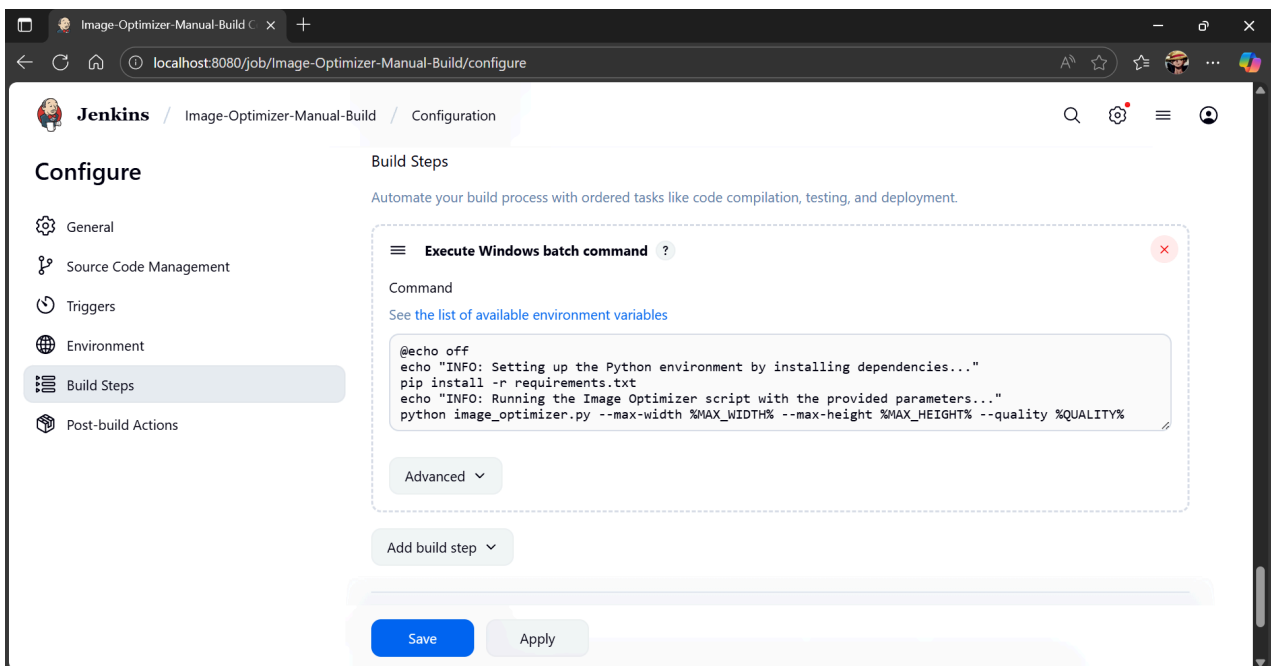
[Save](#) [Apply](#)



The screenshot shows the Jenkins configuration page for a job named 'Image-Optimizer-Manual-Build'. The 'Source Code Management' tab is selected in the left sidebar. The main configuration area includes:

- Branches to build**: A text input field with the value `*/main` and an 'Add Branch' button below it.
- Repository browser**: A dropdown menu currently set to '(Auto)'.
- Additional Behaviours**: An 'Add' button with a dropdown arrow.

At the bottom of the configuration area, there is a 'Save' button and an 'Apply' button.

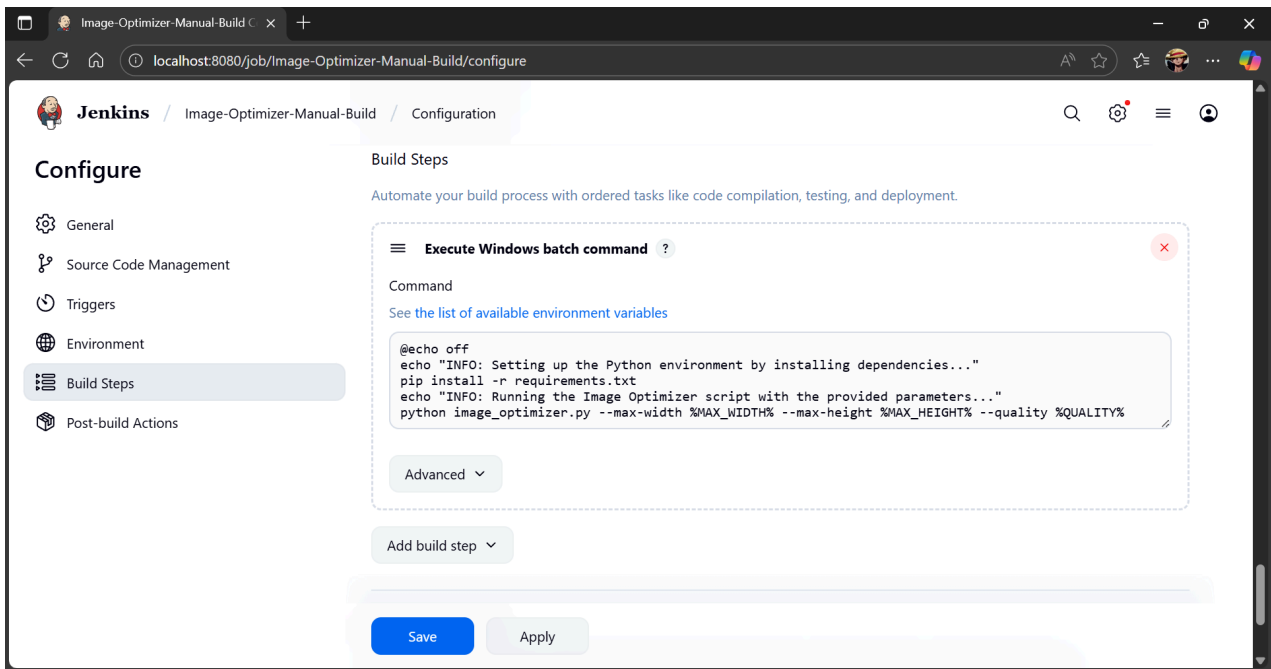


The screenshot shows the same Jenkins configuration page, but with the 'Build Steps' tab selected. The main configuration area includes:

- Build Steps**: A section titled 'Automate your build process with ordered tasks like code compilation, testing, and deployment.' It contains a single step named 'Execute Windows batch command'.
- Command**: A text area containing the following script:

```
@echo off
echo "INFO: Setting up the Python environment by installing dependencies..."
pip install -r requirements.txt
echo "INFO: Running the Image Optimizer script with the provided parameters..."
python image_optimizer.py --max-width %MAX_WIDTH% --max-height %MAX_HEIGHT% --quality %QUALITY%
```
- Advanced**: A button with a dropdown arrow.
- Add build step**: A button with a dropdown arrow.

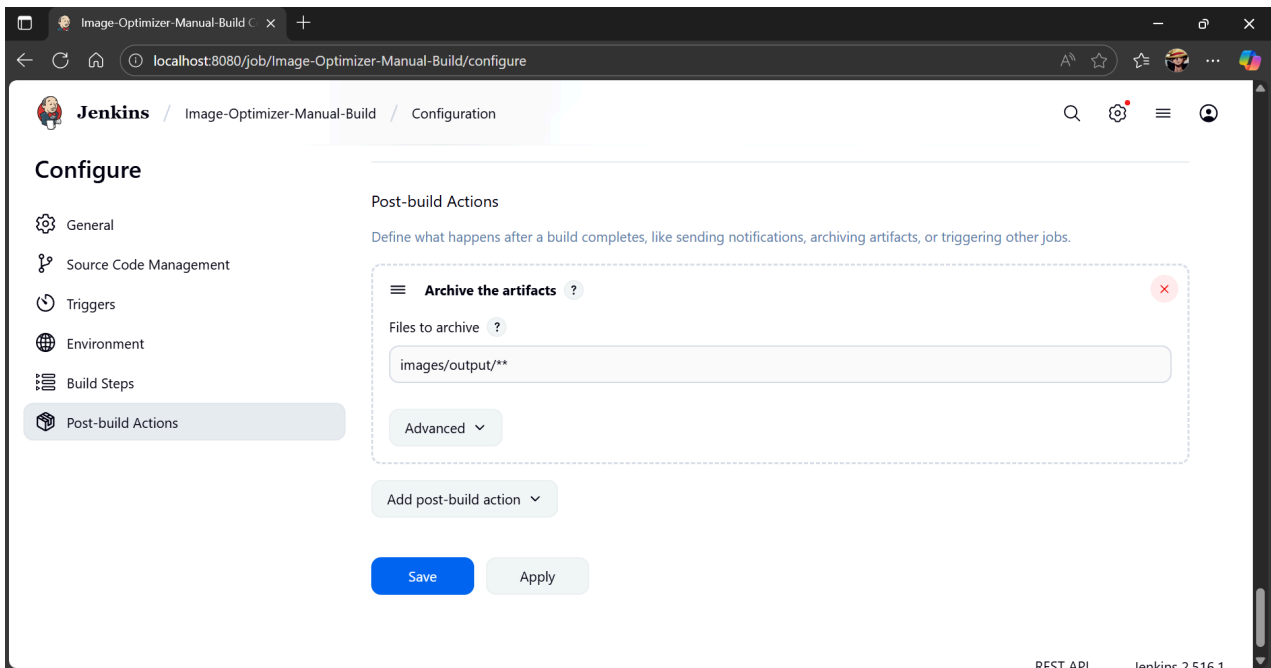
At the bottom of the configuration area, there is a 'Save' button and an 'Apply' button.



The screenshot shows the Jenkins configuration page for a job named 'Image-Optimizer-Manual-Build'. The 'Build Steps' tab is selected in the left sidebar. The main content area is titled 'Build Steps' and includes a description: 'Automate your build process with ordered tasks like code compilation, testing, and deployment.' A single build step is configured with the title 'Execute Windows batch command'. The 'Command' field contains the following text:

```
@echo off
echo "INFO: Setting up the Python environment by installing dependencies..."
pip install -r requirements.txt
echo "INFO: Running the Image Optimizer script with the provided parameters..."
python image_optimizer.py --max-width %MAX_WIDTH% --max-height %MAX_HEIGHT% --quality %QUALITY%
```

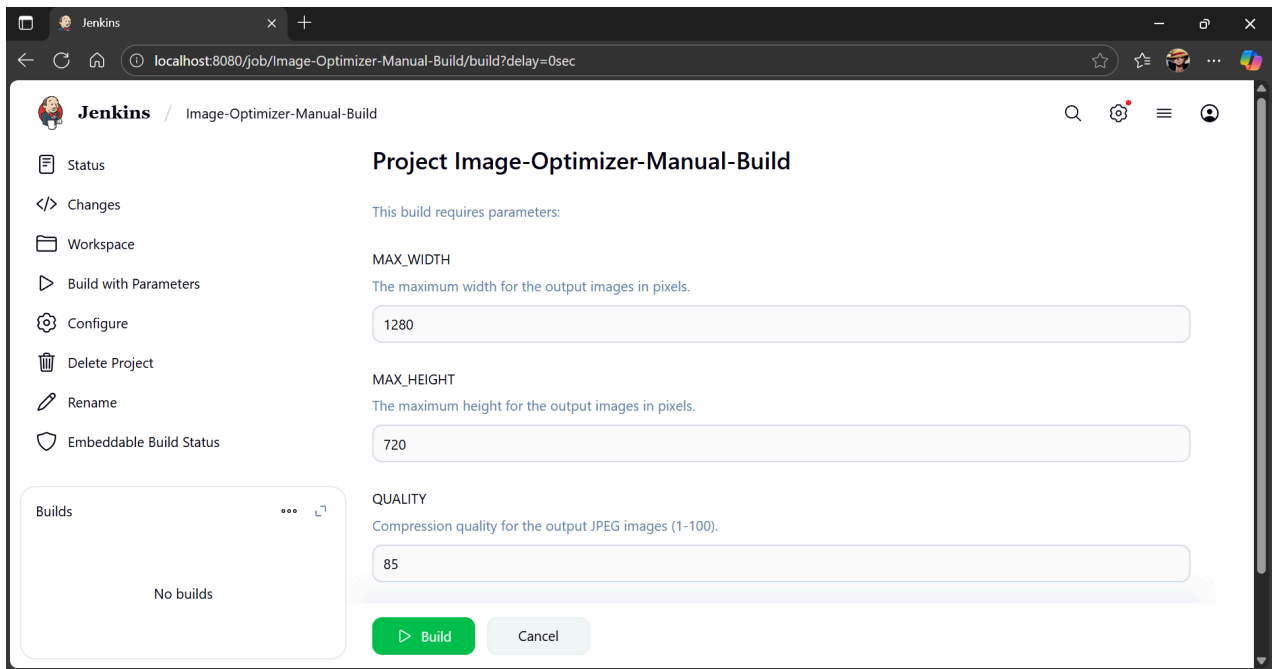
Below the command field is an 'Advanced' dropdown menu. At the bottom of the configuration area are 'Save' and 'Apply' buttons.



The screenshot shows the same Jenkins configuration page, but with the 'Post-build Actions' tab selected in the left sidebar. The main content area is titled 'Post-build Actions' and includes a description: 'Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.' A single post-build action is configured with the title 'Archive the artifacts'. The 'Files to archive' field contains the text:

```
images/output/**
```

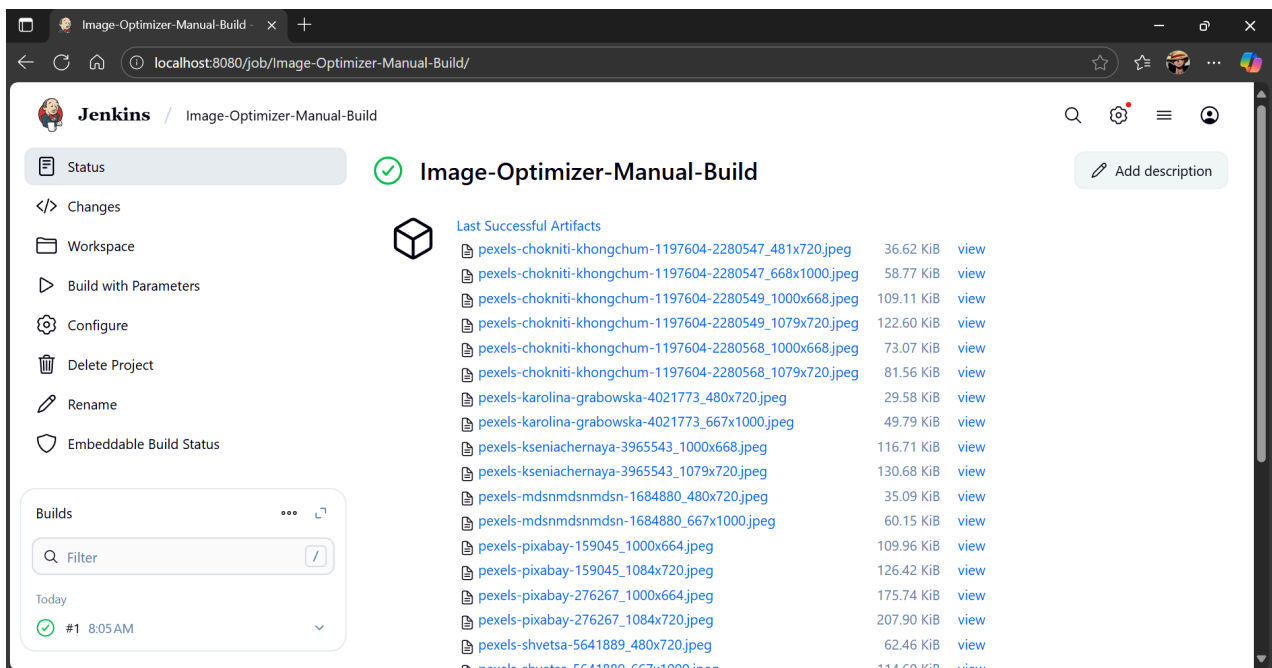
Below the field is an 'Advanced' dropdown menu. At the bottom of the configuration area are 'Save' and 'Apply' buttons. In the bottom right corner, the text 'REST API Jenkins 2.516.1' is visible.



The screenshot shows the Jenkins configuration page for a project named "Image-Optimizer-Manual-Build". The left sidebar contains navigation links: Status, Changes, Workspace, Build with Parameters, Configure, Delete Project, Rename, and Embeddable Build Status. The main area displays the project title and a list of parameters to be configured:

- MAX_WIDTH**: The maximum width for the output images in pixels. Value: 1280.
- MAX_HEIGHT**: The maximum height for the output images in pixels. Value: 720.
- QUALITY**: Compression quality for the output JPEG images (1-100). Value: 85.

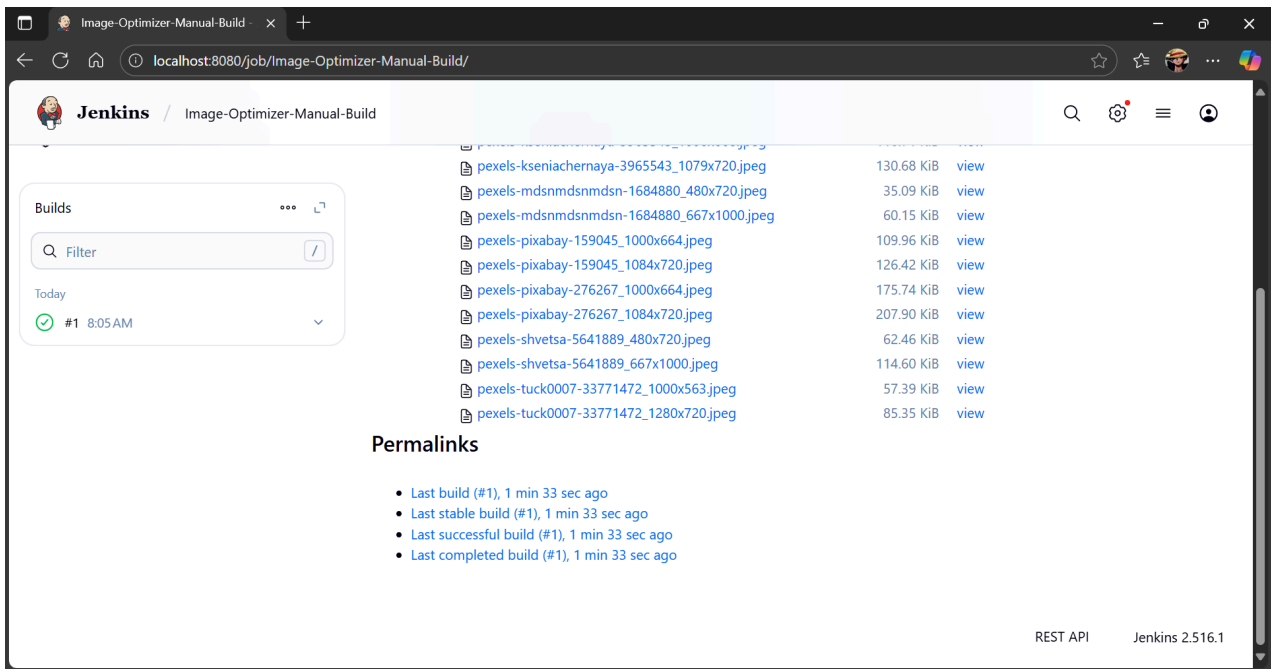
At the bottom, there are "Build" and "Cancel" buttons. A "Builds" section on the left shows "No builds".



The screenshot shows the Jenkins build history page for the "Image-Optimizer-Manual-Build" project. The left sidebar is the same as the configuration page. The main area shows a green checkmark icon and the project name. Below it, a table lists the last successful artifacts:

Artifact Name	Size	Action
pexels-chokniti-khongchum-1197604-2280547_481x720.jpeg	36.62 KiB	view
pexels-chokniti-khongchum-1197604-2280547_668x1000.jpeg	58.77 KiB	view
pexels-chokniti-khongchum-1197604-2280549_1000x668.jpeg	109.11 KiB	view
pexels-chokniti-khongchum-1197604-2280549_1079x720.jpeg	122.60 KiB	view
pexels-chokniti-khongchum-1197604-2280568_1000x668.jpeg	73.07 KiB	view
pexels-chokniti-khongchum-1197604-2280568_1079x720.jpeg	81.56 KiB	view
pexels-karolina-grabowska-4021773_480x720.jpeg	29.58 KiB	view
pexels-karolina-grabowska-4021773_667x1000.jpeg	49.79 KiB	view
pexels-kseniachernaya-3965543_1000x668.jpeg	116.71 KiB	view
pexels-kseniachernaya-3965543_1079x720.jpeg	130.68 KiB	view
pexels-mdsnmdsnmdsn-1684880_480x720.jpeg	35.09 KiB	view
pexels-mdsnmdsnmdsn-1684880_667x1000.jpeg	60.15 KiB	view
pexels-pixabay-159045_1000x664.jpeg	109.96 KiB	view
pexels-pixabay-159045_1084x720.jpeg	126.42 KiB	view
pexels-pixabay-276267_1000x664.jpeg	175.74 KiB	view
pexels-pixabay-276267_1084x720.jpeg	207.90 KiB	view
pexels-shvetsa-5641889_480x720.jpeg	62.46 KiB	view
pexels-shvetsa-5641889_667x1000.jpeg	114.60 KiB	view

Below the table, there is a "Builds" section showing a filter input and a list of builds. The first build is visible: #1 8:05 AM.



Jenkins / Image-Optimizer-Manual-Build

Builds

Filter

Today

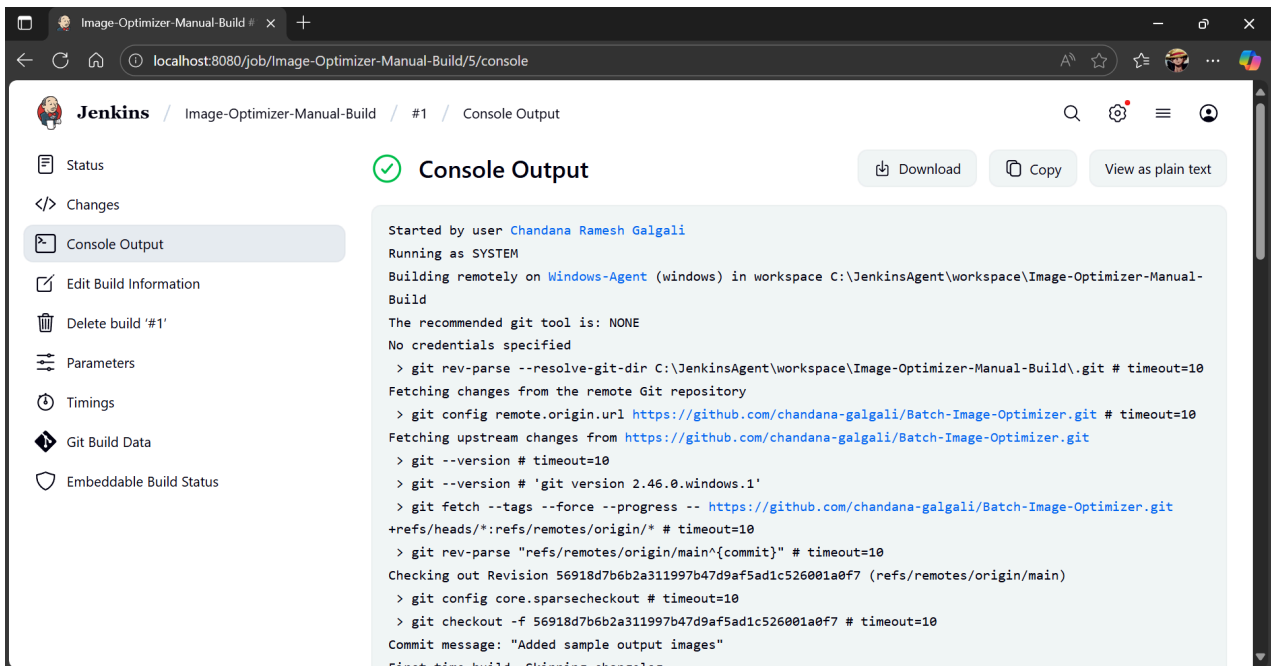
#1 8:05 AM

Build	Time	Size	View
pexels-kseniachernaya-3965543_1079x720.jpeg	130.68 KiB	view	
pexels-mdsnmdsnmdsn-1684880_480x720.jpeg	35.09 KiB	view	
pexels-mdsnmdsnmdsn-1684880_667x1000.jpeg	60.15 KiB	view	
pexels-pixabay-159045_1000x664.jpeg	109.96 KiB	view	
pexels-pixabay-159045_1084x720.jpeg	126.42 KiB	view	
pexels-pixabay-276267_1000x664.jpeg	175.74 KiB	view	
pexels-pixabay-276267_1084x720.jpeg	207.90 KiB	view	
pexels-shvetsa-5641889_480x720.jpeg	62.46 KiB	view	
pexels-shvetsa-5641889_667x1000.jpeg	114.60 KiB	view	
pexels-tuck0007-33771472_1000x563.jpeg	57.39 KiB	view	
pexels-tuck0007-33771472_1280x720.jpeg	85.35 KiB	view	

Permalinks

- Last build (#1), 1 min 33 sec ago
- Last stable build (#1), 1 min 33 sec ago
- Last successful build (#1), 1 min 33 sec ago
- Last completed build (#1), 1 min 33 sec ago

REST API Jenkins 2.516.1



Jenkins / Image-Optimizer-Manual-Build / #1 / Console Output

Status

Changes

Console Output

Edit Build Information

Delete build '#1'

Parameters

Timings

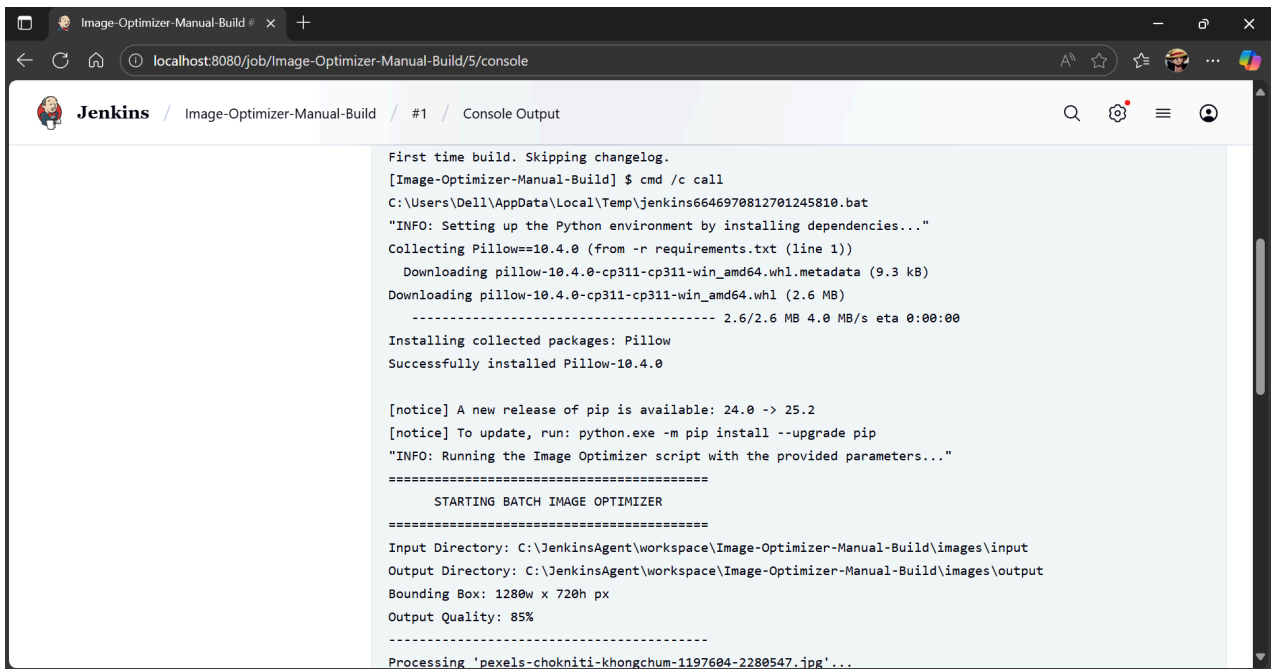
Git Build Data

Embeddable Build Status

Console Output

Download Copy View as plain text

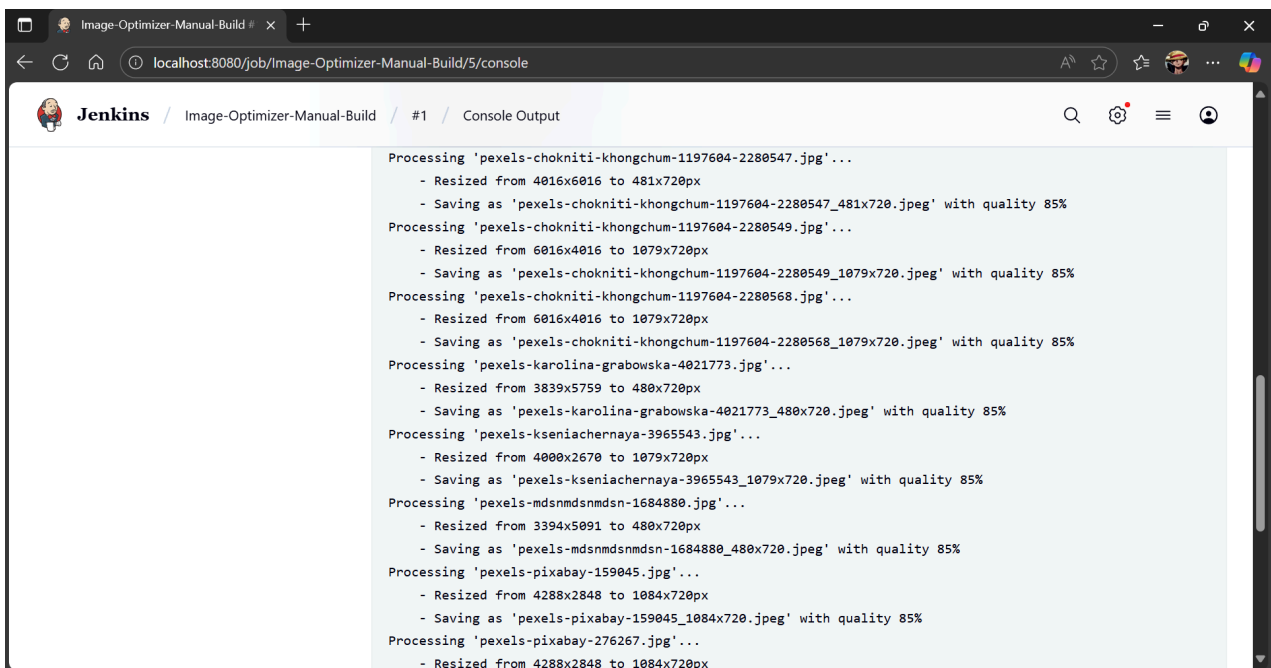
```
Started by user Chandana Ramesh Galgali
Running as SYSTEM
Building remotely on Windows-Agent (windows) in workspace C:\JenkinsAgent\workspace\Image-Optimizer-Manual-Build
The recommended git tool is: NONE
No credentials specified
> git rev-parse --resolve-git-dir C:\JenkinsAgent\workspace\Image-Optimizer-Manual-Build\.git # timeout=10
Fetching changes from the remote Git repository
> git config remote.origin.url https://github.com/chandana-galgali/Batch-Image-Optimizer.git # timeout=10
Fetching upstream changes from https://github.com/chandana-galgali/Batch-Image-Optimizer.git
> git --version # timeout=10
> git --version # 'git version 2.46.0.windows.1'
> git fetch --tags --force --progress -- https://github.com/chandana-galgali/Batch-Image-Optimizer.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> git rev-parse "refs/remotes/origin/main^{commit}" # timeout=10
Checking out Revision 56918d7b6b2a311997b47d9af5ad1c526001a0f7 (refs/remotes/origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f 56918d7b6b2a311997b47d9af5ad1c526001a0f7 # timeout=10
Commit message: "Added sample output images"
First time build. Skipping change log
```



The screenshot shows the Jenkins web interface for a job named 'Image-Optimizer-Manual-Build'. The console output displays the initial build steps, including setting up the Python environment and installing dependencies. The build is currently processing the first image file.

```
First time build. Skipping changelog.
[Image-Optimizer-Manual-Build] $ cmd /c call
C:\Users\Dell\AppData\Local\Temp\jenkins6646970812701245810.bat
"INFO: Setting up the Python environment by installing dependencies..."
Collecting Pillow==10.4.0 (from -r requirements.txt (line 1))
  Downloading pillow-10.4.0-cp311-cp311-win_amd64.whl.metadata (9.3 kB)
  Downloading pillow-10.4.0-cp311-cp311-win_amd64.whl (2.6 MB)
----- 2.6/2.6 MB 4.0 MB/s eta 0:00:00
Installing collected packages: Pillow
Successfully installed Pillow-10.4.0

[notice] A new release of pip is available: 24.0 -> 25.2
[notice] To update, run: python.exe -m pip install --upgrade pip
"INFO: Running the Image Optimizer script with the provided parameters..."
=====
STARTING BATCH IMAGE OPTIMIZER
=====
Input Directory: C:\JenkinsAgent\workspace\Image-Optimizer-Manual-Build\images\input
Output Directory: C:\JenkinsAgent\workspace\Image-Optimizer-Manual-Build\images\output
Bounding Box: 1280w x 720h px
Output Quality: 85%
-----
Processing 'pexels-chokniti-khongchum-1197604-2280547.jpg'...
```



The screenshot shows the Jenkins web interface for the same job, 'Image-Optimizer-Manual-Build'. The console output displays the processing of multiple image files, including resizing and saving them with 85% quality. The build is currently processing the last image file.

```
Processing 'pexels-chokniti-khongchum-1197604-2280547.jpg'...
  - Resized from 4016x6016 to 481x720px
  - Saving as 'pexels-chokniti-khongchum-1197604-2280547_481x720.jpeg' with quality 85%
Processing 'pexels-chokniti-khongchum-1197604-2280549.jpg'...
  - Resized from 6016x4016 to 1079x720px
  - Saving as 'pexels-chokniti-khongchum-1197604-2280549_1079x720.jpeg' with quality 85%
Processing 'pexels-chokniti-khongchum-1197604-2280568.jpg'...
  - Resized from 6016x4016 to 1079x720px
  - Saving as 'pexels-chokniti-khongchum-1197604-2280568_1079x720.jpeg' with quality 85%
Processing 'pexels-karolina-grabowska-4021773.jpg'...
  - Resized from 3839x5759 to 480x720px
  - Saving as 'pexels-karolina-grabowska-4021773_480x720.jpeg' with quality 85%
Processing 'pexels-kseniachernaya-3965543.jpg'...
  - Resized from 4000x2670 to 1079x720px
  - Saving as 'pexels-kseniachernaya-3965543_1079x720.jpeg' with quality 85%
Processing 'pexels-mdsnmdsnmdsn-1684880.jpg'...
  - Resized from 3394x5091 to 480x720px
  - Saving as 'pexels-mdsnmdsnmdsn-1684880_480x720.jpeg' with quality 85%
Processing 'pexels-pixabay-159045.jpg'...
  - Resized from 4288x2848 to 1084x720px
  - Saving as 'pexels-pixabay-159045_1084x720.jpeg' with quality 85%
Processing 'pexels-pixabay-276267.jpg'...
  - Resized from 4288x2848 to 1084x720px
```

The screenshot shows the Jenkins web interface for a job named 'Image-Optimizer-Manual-Build'. The console output displays the following information:

```

- Resized from 4288x2848 to 1084x720px
- Saving as 'pexels-pixabay-159045_1084x720.jpeg' with quality 85%
Processing 'pexels-pixabay-276267.jpg'...
- Resized from 4288x2848 to 1084x720px
- Saving as 'pexels-pixabay-276267_1084x720.jpeg' with quality 85%
Processing 'pexels-shvetsa-5641889.jpg'...
- Resized from 4160x6240 to 480x720px
- Saving as 'pexels-shvetsa-5641889_480x720.jpeg' with quality 85%
Processing 'pexels-tuck0007-33771472.jpg'...
- Resized from 4096x2304 to 1280x720px
- Saving as 'pexels-tuck0007-33771472_1280x720.jpeg' with quality 85%
-----
PROCESS COMPLETE
Successfully processed: 10 images
=====
Archiving artifacts
Finished: SUCCESS
  
```

At the bottom right of the console output area, it says 'REST API' and 'Jenkins 2.516.1'.

Questions:

1. Explain the difference between system configuration and global tool configuration.

Ans:

Aspect	System Configuration	Global Tool Configuration
Purpose	Used to configure overall Jenkins settings, including security, system messages, environment variables, paths, and connected services.	Used to define and configure development tools that Jenkins will use (Java, Git, Maven, etc.).
Scope	Applies to the entire Jenkins server setup and environment.	Applies to the tools available for Jenkins jobs.
Example	Setting up Jenkins URL, managing nodes, email notifications, environment variables.	Setting up JDK installations, Git executable paths, Maven versions.
Effect	Affects how Jenkins as a whole runs and interacts with its environment.	Affects how build jobs find and use necessary tools for execution.

Outcomes: CO3 – Comprehend the effective code building and testing process

Conclusion: (Conclusion to be based on the Results and outcomes achieved)

By successfully executing a Java program using Jenkins with explicit parameterization, we automated the build process and allowed dynamic input values through parameters (**num1** and **num2**). This demonstrated how Jenkins can efficiently integrate code compilation, parameter handling, and execution without manual intervention, ensuring faster and more reliable testing. The outcome shows the importance of continuous integration practices in maintaining code quality and flexibility.

Grade: AA / AB / BB / BC / CC / CD / DD

Signature of faculty in-charge with date

References:

Books/ Journals/ Websites:

1. <https://www.tutorialspoint.com/jenkins/index.htm>
 2. <https://www.guru99.com/create-builds-jenkins-freestyle-project.html>
 3. Effective DevOps: Building a Culture of Collaboration, Affinity, and Tooling at Scale, Jennifer Davis, Ryn Daniels, O'Reilly Media June 2016.
 4. Practical DevOps: Implement DevOps in your organization by effectively building, deploying, testing, and monitoring code, Joakim Verona, Packt Publishing, 2nd Edition, 2016
-